Experiment 4: Autonomous Navigation on the Reconstructed Map

Raw write-up for Chapter 4, Section 4.4

Bolton Athit Davies

Reporting cutoff: 12 November 2026

Purpose of this document. This is the raw, complete write-up of Experiment 4, written before the thesis chapter; Section 4.4 of the final report is extracted from it. The experiment is **partial**. One of its two questions now has data: how well the pose published on a prior map tracks ground truth while the robot drives. The other — whether the planner reaches a goal set without colliding — has none, and its reporting rules are left exactly as they were fixed before any run existed.

4.4.1 Simulation

Status: partial. Seven live runs on 12 November 2026, in three of the four worlds, measure the localisation against ground truth over hand-driven routes of 21–153 m. No goal was commanded in any of them, so the goal-reaching measures of the reporting rules below are still unused and that half of the experiment remains **planned**. The protocol, goal set and success criteria are now fixed in code (`script/exp4_protocol.yaml`, generated by `script/exp4_goals.py`), which is what the previous revision of this document recorded as missing. Mechanisms are analysed separately in `insights/experiment_4_insight/`.

Objective. Establish whether the map of the simulation slot of Experiment 3 is good enough to navigate on: whether the robot can relocalise against a prior it did not build, hold that pose while driving, plan a path through the map, and reach a sequence of goals without colliding with structure the map got wrong. Experiment 3 scores the map as geometry; this experiment scores it as something a planner consumes. The two are not the same question, and the false obstacles Experiment 3 reports are the reason: a map can be 80 % precise and still block the one aisle the robot needs.

Process. `vslam_navigation` exposes the three stages as separate entry points, so the experiment follows them in order: `mapping_sim` builds the prior, `localize_sim` gives a read-only fix against it, and `navigate_sim` adds the planner, the localisation gate and the command interface. The prior is one of the databases already scored in the simulation slot of Experiment 3, so that map quality and navigation outcome are measured on the same artefact. The runs reported here were driven by the operator rather than by the planner; `exp4\_navigate.py` provides both modes (`-drive` and `-run`) and records the same data either way.

Datasets and configuration. The two realistic worlds of the methodology chapter and the closure-free priors built from their mapping recordings: `dataset_allsensor_realistic_map` (U, 148 nodes) and `dataset_allsensor_realistic_01_map` (I, 127 nodes). No bag is replayed: the simulator is driven live, because a navigation outcome depends on the control loop and cannot be reconstructed from a recording. Every run used the configuration frozen in the protocol file: `Vis/EstimationType` 0, which requires matched feature positions to agree in metres rather than merely to reproject, and a 5 s hold

on the back end so that no correction can be latched from the front end's unstable first second. Both settings come from the acceptance sweep of 11–12 November and are recorded in each run's `outcome.json`.

Frames. Ground truth is the Gazebo world frame; the prior, the goals and the published pose are the map frame. The two are reconciled by a closed-form planar fit of every graph node against the mapping run's ground truth, which gives 179.95° and −179.61° of rotation for the U and I priors — the map frame is the spawn, turned around — with median residuals of 0.103 m and 0.055 m. The ground truth reports the model origin and every estimate reports the camera/IMU body, so the 0.338 m lever arm is applied to the ground truth before the fit and before any comparison, and never to recorded data. Two corrections belong on the record here. First, an earlier revision fitted the transform against uncorrected ground truth and reported the resulting 0.337 m residual as confirmation that the fit was right, on the grounds that it equalled the lever arm. That was the bias, not the evidence: a rigid transform cannot absorb an offset that rotates with the body, so it absorbed part of it as translation and left the rest as residual. Applying the lever arm before the fit drops the residual to 0.103 m and 0.055 m.

Second, and larger, the same revision applied the lever arm to ground truth in the *comparison* as well, which was wrong and inflated every error this slot reported by about a third of a metre. The recorder samples `map` → `base_link`, and the published `base_link` sits where the model origin is, so ground truth needs no correction to meet it. Decomposing the error along and across the direction of travel is what exposed it: with the lever arm applied, the along-track mean was −0.28–−0.38 m in all seven runs while the cross-track mean sat near zero — a constant displacement in the direction of travel, of the lever arm's length, in the same direction every time, and not scaling with speed, so not a latency either. Without it the along-track mean is −0.16–0.04 m and changes sign between runs. The medians fell from 0.36–0.62 m to 0.10–0.36 m accordingly. The lever arm still belongs in the alignment, which fits body-frame node poses; the two are different quantities and their frames legitimately differ.

Reporting rules, goal-reaching. Fixed before any run exists, so that the result cannot be chosen after the fact, and left unchanged here. A goal counts as reached when the base frame is within 0.25 m and 10° of it and the planner reports success. Per run the measures are: the share of goals reached, the path length against the shortest collision-free path in the ground-truth occupancy grid, the number of collisions with true structure, the number of Nav2 recovery behaviours triggered, the share of the run with a valid localisation fix, and the wall time to complete the goal set. A run that loses localisation and does not recover within 30 s is reported as failed, not excluded.

Reporting rules, pose tracking. Defined after these data existed, and reported as the weaker claim it is. Error is the planar distance between the published `map` → `base_link` transform, sampled at 2 Hz, and the lever-arm-corrected ground truth interpolated to the same instants. That transform is used rather than `/localization_pose` because it is what RViz draws and Nav2 consumes, and it includes the front end's dead reckoning between accepted fixes. The error is reported as a median, a 95th percentile, a maximum, the median over each half of the run, and split into the length of the mean error vector (offset) and the standard deviation about it (scatter). A pose step over 1 m between consecutive samples is counted as a jump, excluding the seed transition and the acquisition snap. Every run is reported on its own, as in the other slots.

Results. Table 4.1 gives the seven runs and figures 4.2 to 4.5 draw four of them. The pose tracks ground truth to a median of 0.10–0.36 m, with a median across runs of 0.162 m, and the error is bounded rather than growing: in six of the seven runs the median over the first and second half of the run agree. Minimum ground-truth clearance to a static model never fell below 0.58 m, so no run collided.

Table 4.1: Pose tracking against ground truth, 12 November 2026. Route shapes were driven by hand: straight, approximately square, approximately a figure-of-eight. "Over 1 m" is the longest unbroken stretch with the error above a metre and "ended" the median over the last tenth of the run, i.e. whether it came back. Offset and scatter, reported in the text, are 0.07–0.27 m and 0.04–0.24 m respectively. Blind is the share of localisation poses carrying RTAB-Map's no-information covariance. Ground truth is compared without a lever arm, for the reason given under Frames.

World	Route	Driven (m)	Dur. (s)	Med. (m)	p95 (m)	Max (m)	Over 1 m (s)	Ended (m)	Blind	Jumps
U static	straight	24.6	25	0.162	0.226	0.24	0.0	0.033	0.045	0
U static	square	109.2	86	0.188	6.355	6.80	36.0	0.106	0.016	4
U static	figure-8	152.0	124	0.098	3.750	3.99	23.5	0.120	0.013	3
I static	straight	20.9	25	0.097	0.598	0.61	0.0	0.084	0.158	1
I static	square	66.8	57	0.361	0.631	0.69	0.0	0.113	0.068	2
I static	figure-8	123.7	122	0.136	0.749	0.93	0.0	0.117	0.047	1
U dynamic	straight	26.0	30	0.298	1.077	1.17	2.5	0.066	0.103	1



Figure 4.1: Every run on one axis: planar error against distance driven since the first fix, logarithmic. Distance rather than time, so that runs driven at different speeds compare. The two grey lines are the rms accuracy Experiment 3 measures for these priors' own graphs, 0.10–0.40 m: most of the time most runs sit between them, which is the bound this slot reports. The two shelves are the excursions of table 4.1 — flat, not ramped, so the pose steps to a wrong place and holds it rather than drifting away. Both end abruptly when a correct fix lands.

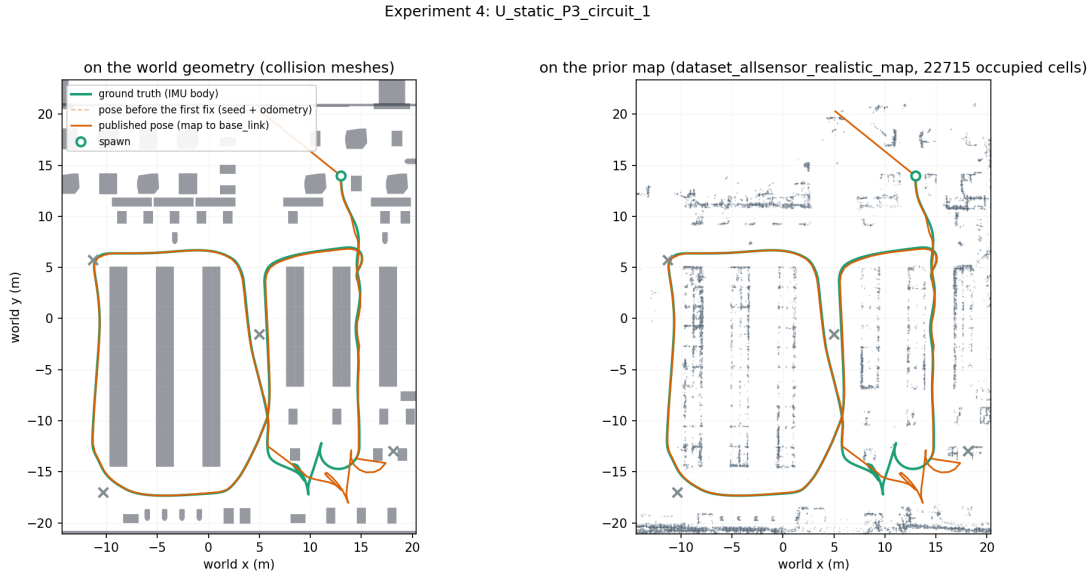


Figure 4.2: `U_static`, figure-of-eight, 152 m. Left: the published pose against ground truth on the world's collision-mesh footprints, the same underlay the chapter 3 scene figures use. Right: the same two tracks on the prior's exported occupancy grid, brought into the world frame by the fit of section 4.4.1. The grid's shelf rows landing on the mesh footprints is the alignment confirming itself; the thick band at the right wall against the mesh's thin line is the wall displacement Experiment 3 reports. The dashed segment is the pose before the first fix, which is the seed composed with raw odometry and is not a tracking error. Median error 0.098 m.

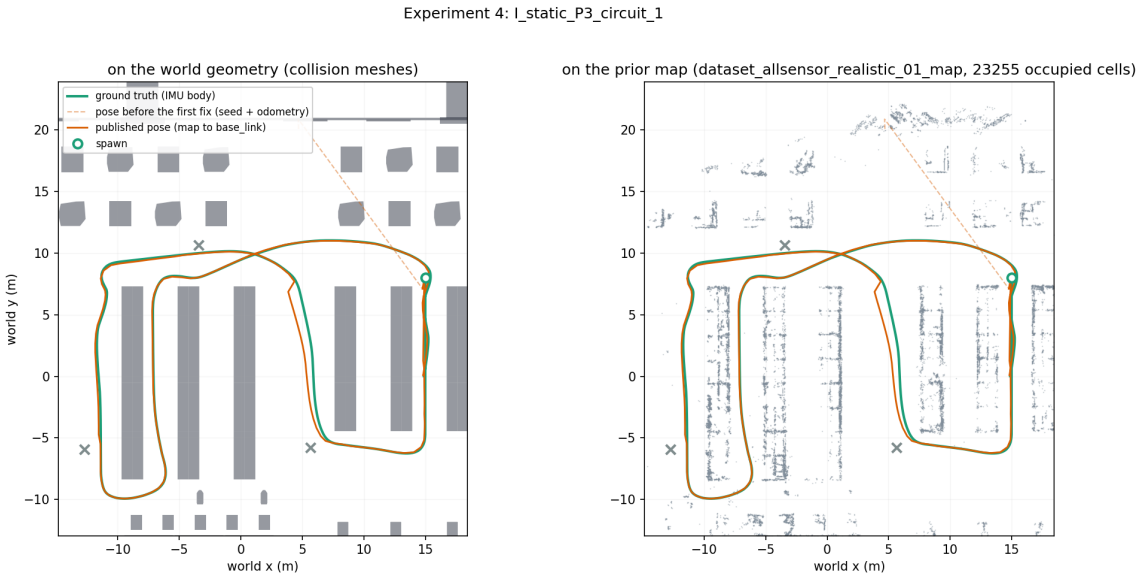


Figure 4.3: `I_static`, figure-of-eight, 124 m: the cleanest run of the seven, median 0.136 m and maximum 0.93 m, on the second layout and its own prior (`dataset_allsensor_realistic_01_map`, 23 255 occupied cells). The layouts differ in aisle arrangement, so this is not a repeat of figure 4.2 in a different place: the route crosses its own path twice and the pose holds through both crossings, which is where a prior that aliases between aisles would show it. Conventions as figure 4.2.

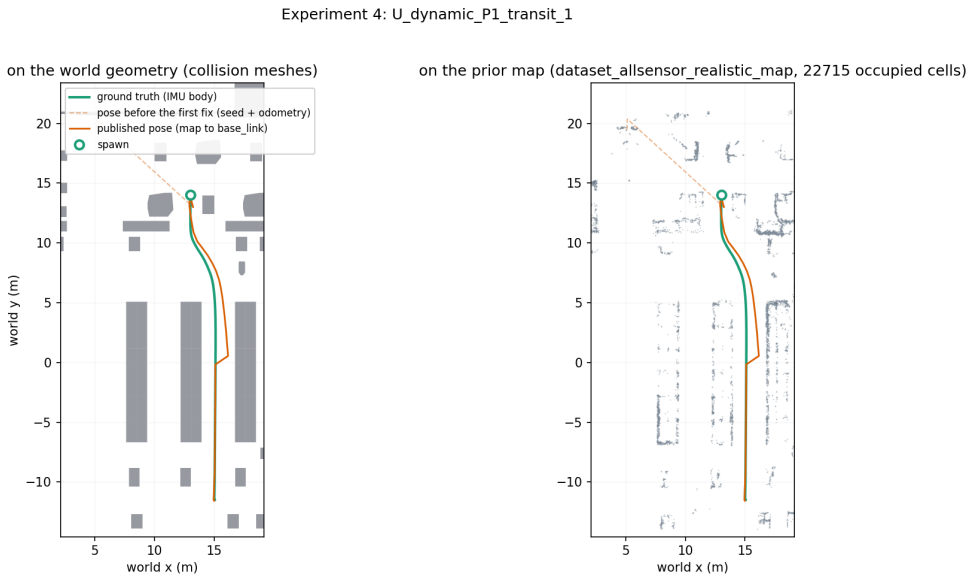


Figure 4.4: `U_dynamic`, straight, 26 m: the worst run of the seven on both offset (0.274 m) and scatter (0.240 m), and the only one with traffic. The published pose runs beside ground truth down the aisle rather than diverging from it, which is what a displacement rather than a drift looks like. Conventions as figure 4.2.

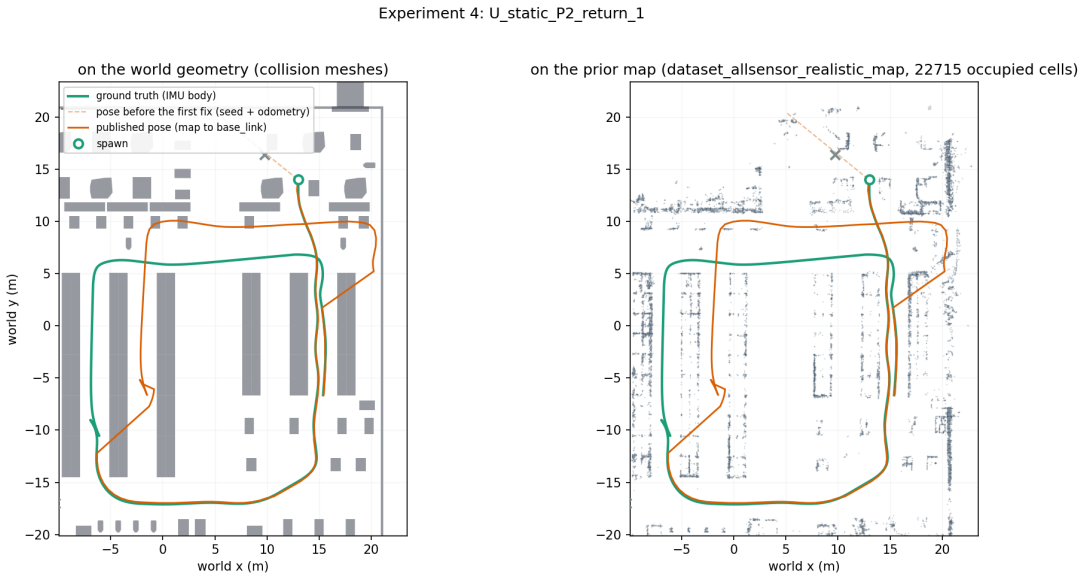


Figure 4.5: `U_static`, approximately square, 109 m: the one run of the seven that lost the pose. From about $y = -5$ the published pose leaves the aisle and cuts diagonally through three shelf rows while ground truth follows the aisle, which is the 6.09 m second-half median of table 4.1 seen directly. A pose that passes through shelving is wrong in a way no error statistic states as plainly. Conventions as figure 4.2.

Heading. Measured in the insight analysis (`insights/experiment_4_insight/`) and reported here because the slot otherwise never looks at orientation. Median heading error is below 1.2° in all six static runs and the 95th percentile below 3.6° in five of them, which is not a limiting quantity for a 0.6 m wheelbase in an aisle: a degree converts to 1.7 cm of cross-track error per metre driven. The dynamic run is the exception, at 4.85° with a 3.35° bias — four to thirteen times every static run. Traffic therefore costs heading about tenfold where it costs position about threefold, so the position figures understate it. Heading error does not rise with turn rate in any run.

Insight: the pose is as good as the map it is localising against. The error decomposes into a constant offset of 0.07–0.27 m and scatter of 0.04–0.24 m about it, and almost all of it is across the direction of travel rather than along it: once the frame error above is removed, the along-track mean is within 0.16 m of zero in every run and changes sign between them. Cross-track error is the component that matters for a corridor, and it is the smaller one.

The magnitude is the reading. Experiment 3 measures the graph of these same priors as accurate to 0.10–0.40 m rms over the 365–381 m build routes. The pose tracking that prior is accurate to 0.10–0.36 m. The localiser is therefore reproducing its map to within the map's own accuracy, and nothing in these runs shows it doing worse than the artefact it is given. That is a bound, not a coincidence: an estimate cannot be closer to the world than the map it is matched against, so this slot's accuracy is *map-limited rather than estimator-limited*, and the improvement to chase is a better prior rather than a better filter.

The previous revision drew a stronger conclusion from the same data — that the offset was route-dependent, 0.32–0.59 m on a single aisle against 0.02–0.14 m over the building, and matched Experiment 3's half-metre wall displacement. That pattern was the frame error, which is larger on a straight run where the heading barely changes. It does not survive the correction and is withdrawn.

Insight: the initialisation transient damages both layers. Experiment 3 attributes the blocking false obstacles of three of the four realistic maps to phantom cells laid by the first five graph nodes (63–96 % of them), and names discarding those cells, or starting the map after the inertial transient has settled, as the one correction its data predict would help. The same transient was found independently in this slot, one layer up: the front end's orientation is unstable for about a second after it starts, and a back end that accepts a fix during that window latches a correction 73–76° wrong and holds it for roughly 16 s. Holding the back end back for 5 s removed it, confirmed over three runs by the collapse of the largest correction yaw from 110–177° to 25°. Neither `wait_for_imu_init` nor the pose gate's `init_ignore_s` prevents it: the first tests inertial validity, which is already true through the unstable window, and the second only withholds the pose from Nav2 after the correction has been formed. One transient, two symptoms, and the same prescription in both places — do not consume the first second.

Excursions, and that every one of them ended. Two runs left the map for tens of seconds. The U static square route was over a metre from ground truth for 36 s, reaching 6.8 m, and the U static figure-of-eight for 23.5 s, reaching 4.0 m. Both then recovered: the last tenth of each run sits at 0.106 m and 0.120 m. The U dynamic run had a shorter one, 2.5 s. The four remaining runs never crossed a metre at all. Figure 4.1 shows all seven on one axis, and the two long excursions as the flat shelves they are — the pose does not drift away and come back, it steps off, holds a wrong place, and steps back when a correct fix lands.

The two are not the same fault, which the corrections show and the error curves do not. On the figure-of-eight the correction's yaw moves from −0.5° to 11.0° and back, heading error inside the stretch is 11.0° against 0.4° outside it, and heading and position error correlate at $r = 0.97$: the robot was placed facing the wrong way, and driving converted the angle into displacement. On the square the correction's yaw barely moves while its translation shifts over 5 m: the robot was placed in the wrong aisle at the right heading. These are the warehouse's two symmetries, the aisle pitch and the orientation of a bay, and a defence aimed at one does not catch the other — the same split the November acceptance sweep found

among individual rejected fixes.

Two earlier claims are withdrawn. The square route was described as having "lost the pose" and "not recovered before the run ended"; it recovered 9 s before the end, to 0.011 m on the final sample. The figure-of-eight was reported as clean, because its excursion sits inside the first half and a half-by-half median hid it. Both errors came from summarising with medians and a maximum and no measure of duration, which is why the table now carries one.

Availability is the weaker column. The I static straight run carried no information for 15.8 % of its poses and the U dynamic run for 10.3 %, against a criterion of 10 %. Accuracy in both was ordinary when a fix existed, so the difficulty is holding the fix rather than obtaining one.

Insight: over these distances the prior supplies a datum, not accuracy. This narrows the objective stated above, and is the claim the data support. Unaided ORB-SLAM3, rigidly aligned onto ground truth, drifts by a median of 0.045–0.345 m over these 21–152 m routes, never exceeds 0.76 m at any instant, and ends each run between 0.03 m and 0.60 m from truth. Run by run the medians split three to three with one tie against the localised pose, so localising on the map does not improve local accuracy here — and it introduces excursions an order of magnitude larger than the front end produced alone, 6.80 m and 3.99 m against a front-end worst of 0.76 m.

What the prior does supply is the thing that comparison gives the front end for free: an absolute, repeatable datum in a frame that outlives the run, which is what lets a goal in map coordinates mean the same place twice. Over these route lengths that is the whole of its contribution, bought at two multi-second excursions in seven runs. The prediction worth testing is that it begins to pay for itself in accuracy where front-end drift exceeds the localisation error, beyond 152 m; the 365–381 m mapping routes should already reach that.

Traffic. The one dynamic run is the worst of the seven on both offset (0.588 m) and scatter (0.247 m, about double the static runs) and sat unlocalised 10.3 % of the time. One run decides nothing, but it points the same way as Experiment 1's finding that traffic degrades the estimate without breaking tracking.

What these data cannot answer. Every cell is a single run, and the acceptance sweep of 11–12 November established that this stack's run-to-run spread over identical settings is as wide as most differences between settings, so no comparison between worlds, layouts or routes in table 4.1 is evidence. The routes were driven by hand and differ in shape and length, so route is a confound rather than a factor. No goal was commanded, so nothing here speaks to planning, to the false-obstacle question that motivates the experiment, or to recovery behaviour. One dynamic world is sampled and one is not.

Next action. Collect the dynamic worlds and reps 2 and 3 of every cell, then run the goal set with `-run` so the goal-reaching rules above can be applied. The narrow question still outstanding is Experiment 3's: does the 3.8–10.2 % false-obstacle rate block a route, or does the planner drive around it.

4.4.2 Every run

The three runs not drawn above, for completeness; conventions as figure 4.2.

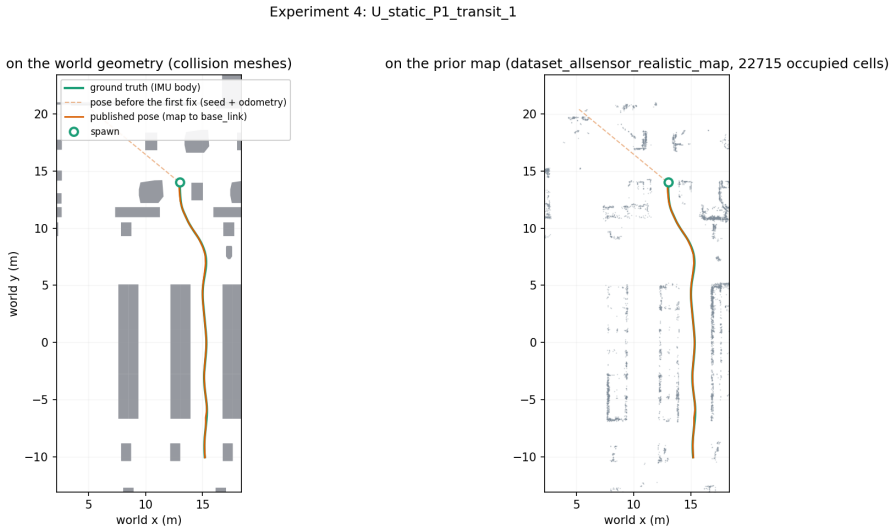


Figure 4.6: `U_static`, straight, 24.6 m. The two tracks are indistinguishable at this scale, which is the point: the median error is 0.162 m and the run never crossed a metre. It is also the figure that exposed the lever-arm error — a 0.47 m offset was being reported for a run whose tracks visibly coincide, which is only possible if the offset lies along the direction of travel, where a path plot cannot show it.

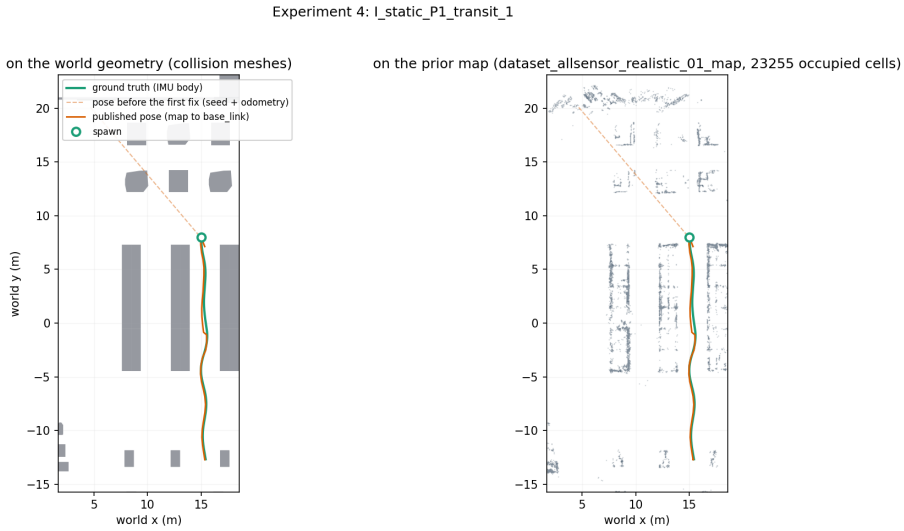


Figure 4.7: `I_static`, straight, 20.9 m: the most accurate run of the seven by median (0.097 m) and the least available, with 15.8 % of its poses carrying no information. Accuracy and availability are separate properties and this run holds the extreme of both.

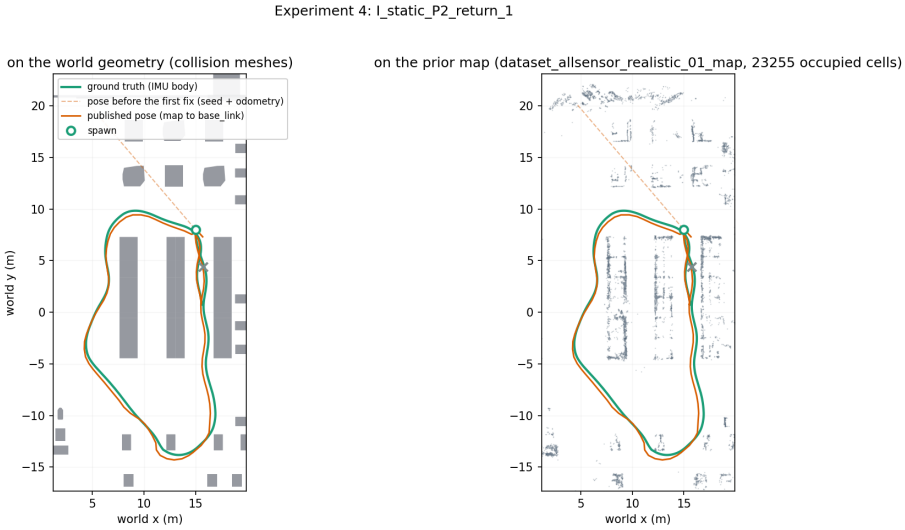


Figure 4.8: `I_static`, approximately square, 66.8 m: the largest median of the seven at 0.361 m, with no excursion over a metre and an offset of 0.204 m. A consistently displaced pose rather than an unstable one, and the case that sets the upper end of the 0.10–0.36 m range.

4.4.3 Real robot

Status: planned. No real-robot navigation run has been made, and the real rig has no map to navigate on: the dense-map slot of the real-robot slot of Experiment 3 is itself planned, because no RTAB-Map database has been built from real imagery and no reference geometry of the site exists to score one against.

Objective. As in simulation, on the real rig.

Next action. Blocked behind the real-robot half of Experiment 3: a map must exist before navigation on it can be scored. The 5 s back end hold that works in simulation is a fixed guess at the length of the front end's unstable window and should not be carried to the rig unchanged; the transferable form is an orientation-stability condition in the front end itself.